

WASP-12b

R-0392

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_230129 · created 2026-07-19T00:37:16+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459973.71182 +/- 0.00064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1528 +/- 0.0069
Transit depth (Rp/Rs)^2	2.33 +/- 0.21 [%]
Orbital Inclination (inc)	86.77 +/- 0.79
Airmass coefficient 1 (a1)	1.2831 +/- 0.0099
Airmass coefficient 2 (a2)	-0.231 +/- 0.029
Scatter in the residuals of the lightcurve fit is	0.77 %
Best Comparison Star	None
Optimal Aperture	4.24
Optimal Annulus	13.43
Transit Duration (day)	0.1337 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1528 $\pm$ 0.0069 vs 0.1178 (deviation 5.07 σ)
Duration fit vs published	0.1337 d vs 0.125 d (deviation 7.91 σ)
Mid-time O-C	-3.3 min
Depth SNR	22.1
Residual scatter	0.77 %

Light curve

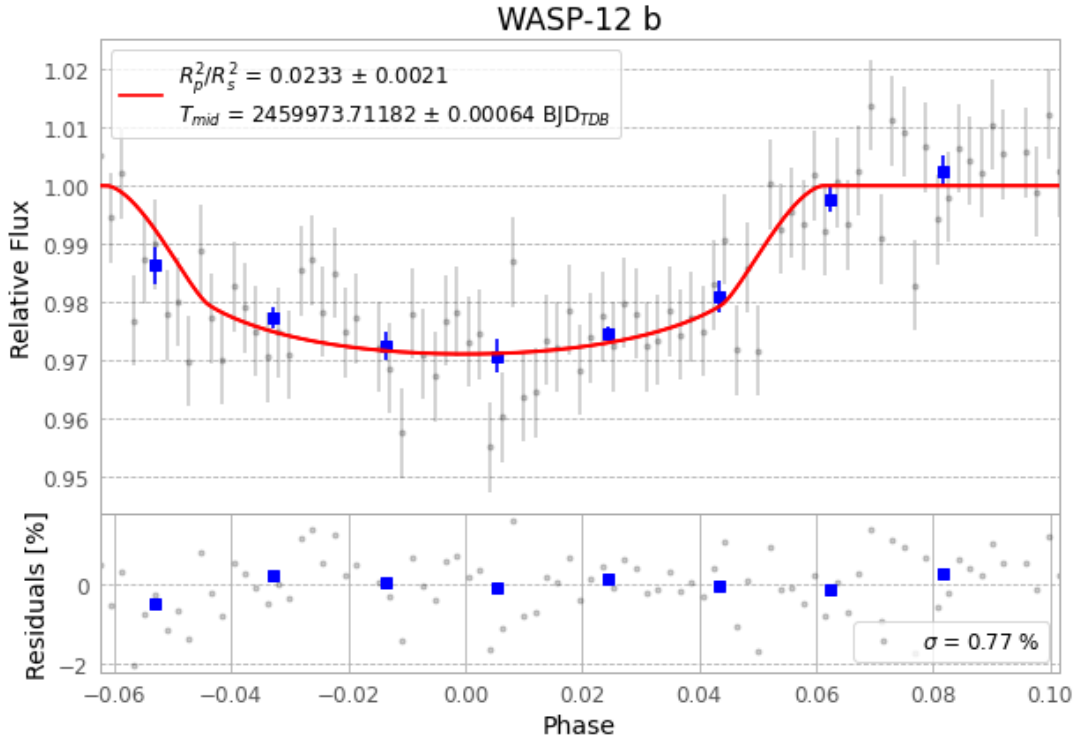


Figure 1 — FinalLightCurve_WASP-12 b_2023-01-28.png (SHA-256 9005bedf9224094d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [431, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3dd6acd2cc38c134...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

87 FITS frames (57 MB), WASP-12230129031816.FITS → WASP-12230129073615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230129.log (5140cd44b3e7...), FinalLightCurve_WASP-12 b_2023-01-28.png (9005bedf9224...), FinalLightCurve_WASP-12 b_2023-01-28.csv (4df41c8c1ddc...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 00:37Z.